

SDS-PAGE

Overview

To separate proteins based on their molecular weight using polyacrylamide gel electrophoresis.

Materials

Component	Stacker (top gel)	Resolver (bottom gel)
Resolver A	-	4 mL × n
Resolver B	-	4 mL × n
Stacker A	1.5 mL × n	-
Stacker B	1.5 mL × n	-
Total Volume	3 mL × n	8 mL × n
TEMED	3 µL × n	4 µL × n
10% APS	15 µL × n	40 µL × n

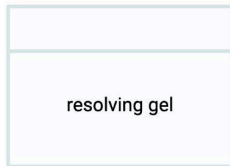
A. Procedure

1. Mix equal amounts of Resolver A and Resolver B solutions to prepare resolving gel acrylamide solution.
2. Add 4 µL × n of TEMED and 40 µL × n of freshly prepared 10% APS to the resolving solution; mix thoroughly.
3. Prepare the stacking gel by combining equal amounts of stacker A and B solutions (1.5 mL × n each).
4. Add 3 µL × n of TEMED and 15 µL × n of freshly made 10% APS to the stacker solution; mix thoroughly.
5. Carefully pipet the stacking solution straight down the middle of the cassette up to the top of the short plates, pouring smoothly so it does not mix with the resolving gel layer.

6. Place the comb into the cassette alignment slots slowly to avoid trapping bubbles beneath the teeth.
7. Leave the gel undisturbed for 30–45 minutes to complete polymerization before running.
8. Combine 5–15 μL of protein supernatant with 6X Laemmli buffer (stored at 4°C) to bring the total sample volume to 20 μL .
9. Combine 5 μL of protein ladder (stored at -20°C) with 1 μL of 6X Laemmli buffer.
10. Heat all prepared tubes in a heat block at 95°C for 10 minutes.
11. Spin down the tubes briefly to collect liquid prior to loading.
12. Load 2–3 μL of prepared protein ladder into both the first and last wells. Fill the middle wells with your protein samples. Fill any unused wells with 1X Laemmli buffer to prevent adjacent sample bands from spreading.
13. Run the SDS-PAGE gel at a constant 200 V for 10 minutes until the sample tracking dye completely leaves the stacking layer and enters the bottom gel.
14. Lower the voltage to 120 V (constant) and continue running for approximately 45 to 60 minutes until the dye front approaches the gel bottom.

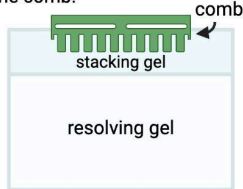
SDS Page

- 1 Prepare the resolving gel in the Polyacrylamide Gel Electrophoresis.



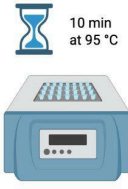
Leave undisturbed
10-20 min for
polymerization

- 2 Add the stacking gel until it overflows and then insert the comb.



Leave undisturbed
30-45 min for
polymerization

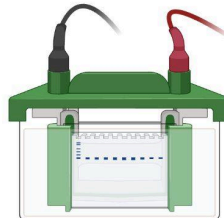
- 3 While waiting for the gel to polymerize, prepare the sample.



- 4 Load the samples into the wells when the gel finishes polymerizing.



- 5 Electrophorese for 10 minutes at 200 V, then 45-60 minutes at 120 V (until the dye reaches the bottom of the gel).



Coomassie Blue Staining

A. Materials

Coomassie Blue Solution (50 mL)		
Material	Percent Composition (%)	Volume (mL)
CBR-250	0.05	0.025
Methanol	50	25
Acetic acid	10	5
ddH ₂ O	39.95	19.975

Coomassie Blue Destain Buffer (50 mL)

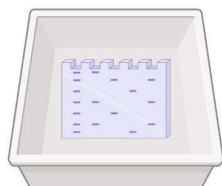
Material	Percent Composition (%)	Volume (mL)
Methanol	30	15
Acetic acid	10	5
ddH ₂ O	60	30

B. Procedure

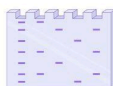
1. Prepare the 50 mL Staining and Destain solutions according to the table above.
2. Place the gel in a plastic container and wash with ddH₂O on an orbital shaker 3 times (5 minutes per wash) to remove residual SDS.
3. Use 0.05% Coomassie blue R250 for 1 hr on the shaker to stain the gel.
4. Replace the stain with Coomassie Blue Destain Buffer and shake for at least 1 hour until clear protein bands appear against a transparent background.

Coomassie Blue Staining

- 1 Wash the gel with ddH₂O 3 times for 5 minutes each time.



- 2 Stain the gel with 0.05% Coomassie Blue R250 on the shaker at 50 rpm for 1 hour.



- 3 Soak the gel in Coomassie Blue Destain Buffer for at least 1 hour and until clear.

